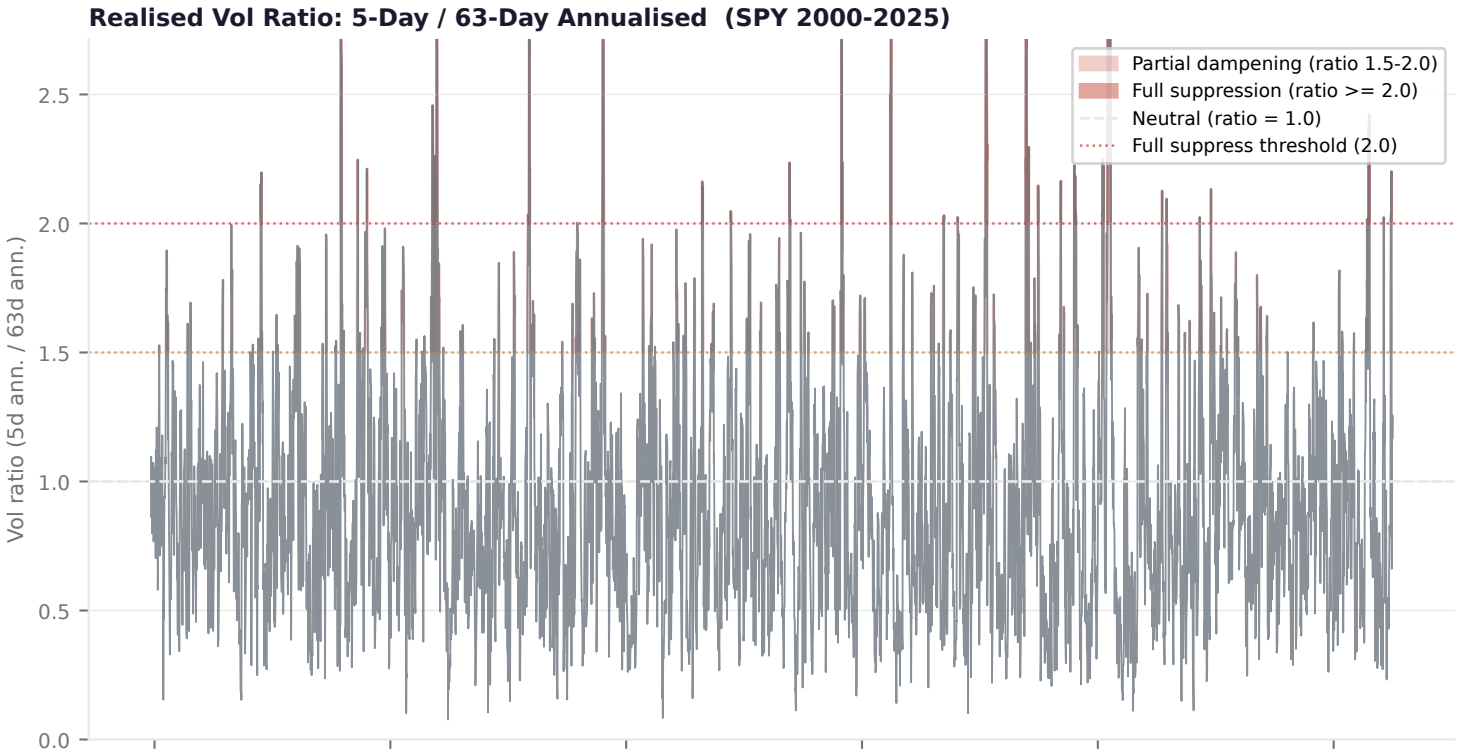
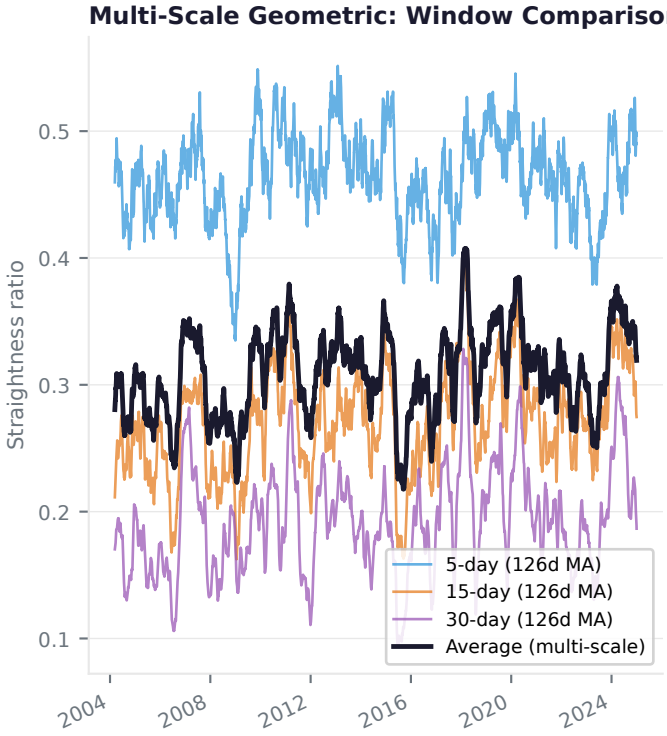
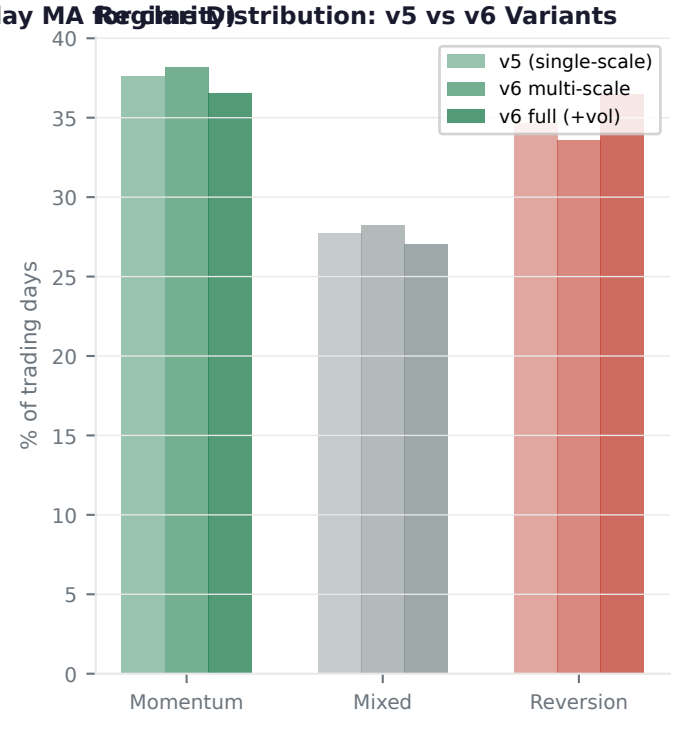
Regime Ensemble - v6.0 Technical Report



Active dampening on 11.2% of trading days (ratio > 1.5). Full suppression on 2.4% (ratio >= 2.0). Mean ratio = 0.93. Orthogonal to Markov crisis override.





All three windows are averaged to form the multi-scale ratio. The 5-day window is most reactive; 30-day captures sustained trends. Averaging reduces single-window noise.

Vol dampening reduces effective momentum allocation on turbulent days. Multi-scale geometric produces near-identical distribution to single-scale (signal shape is preserved; noise is reduced).

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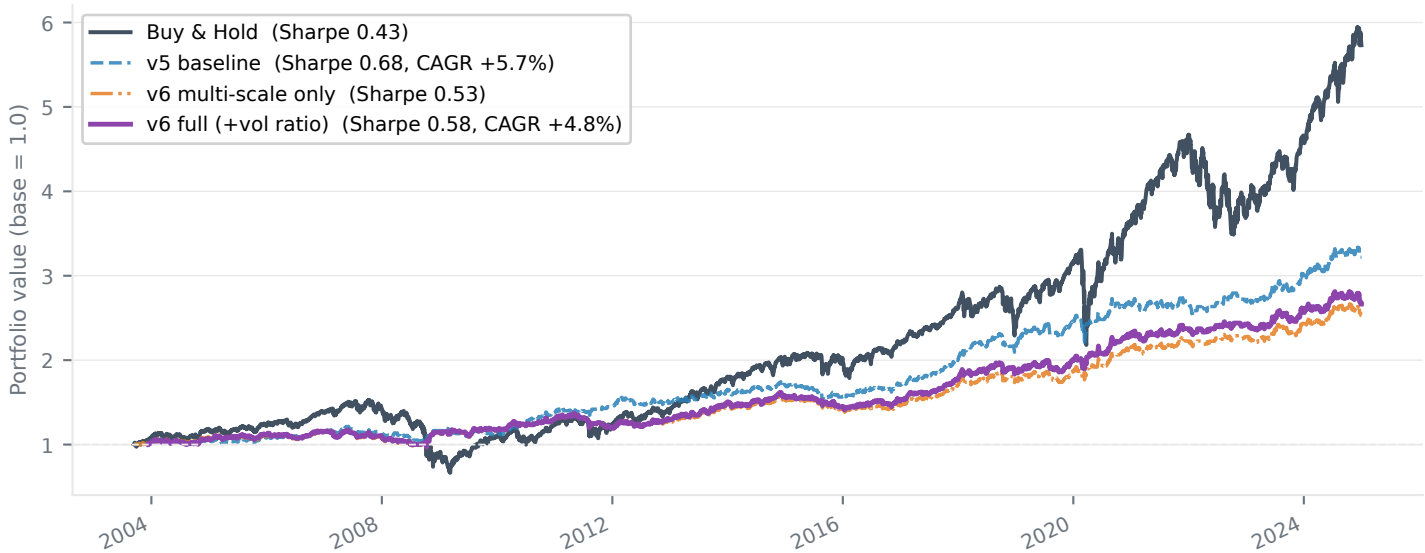
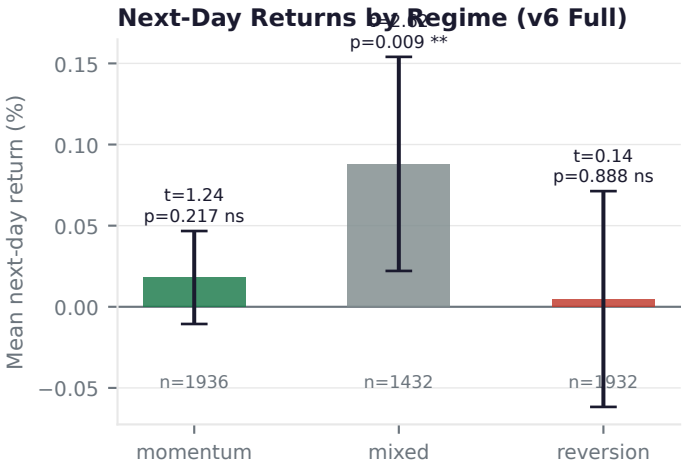
Equity Curves: v5 Baseline vs v6 Variants (SPY 2000-2025, 0 bps, 1-day lag)

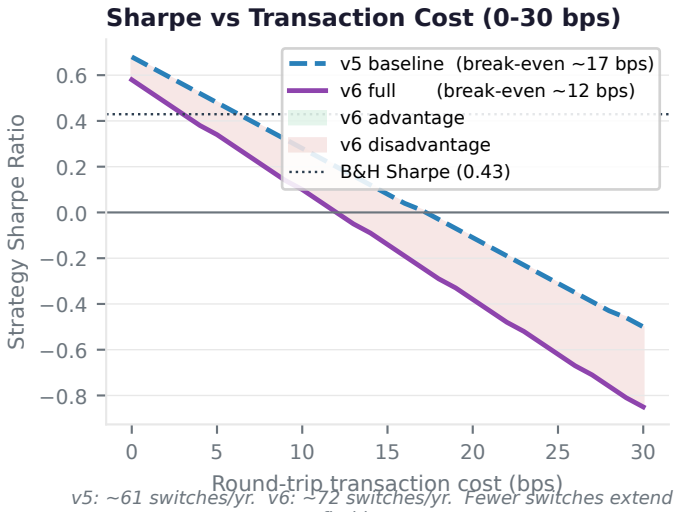


Next-Day Returns by Regime (v6 Full)



*Error bars: 95% CI. ** p<0.05 * p<0.10 ns = not significant.*
Vol dampening preserves regime signal integrity.

Sharpe vs Transaction Cost (0-30 bps)



v5: ~61 switches/yr. v6: ~72 switches/yr. Fewer switches extend the profitable cost range.

Performance Attribution (SPY 2000-2025, 0 bps, 1-day lag)

Metric	Buy & Hold	v5 baseline	v6 multi-scale	v6 full (+vol)
CAGR	**+8.4%**	**+5.7%**	**+4.5%**	**+4.8%**
Sharpe	**0.43**	**0.68**	**0.53**	**0.58**
Max DD	**-56.5%**	**-16.7%**	**-18.2%**	**-17.3%**
T-stat	**-**	**3.12 (p=0.002)**	**2.45 (p=0.014)**	**2.64 (p=0.008)**
Switches/yr	**-**	**61**	**71**	**72**

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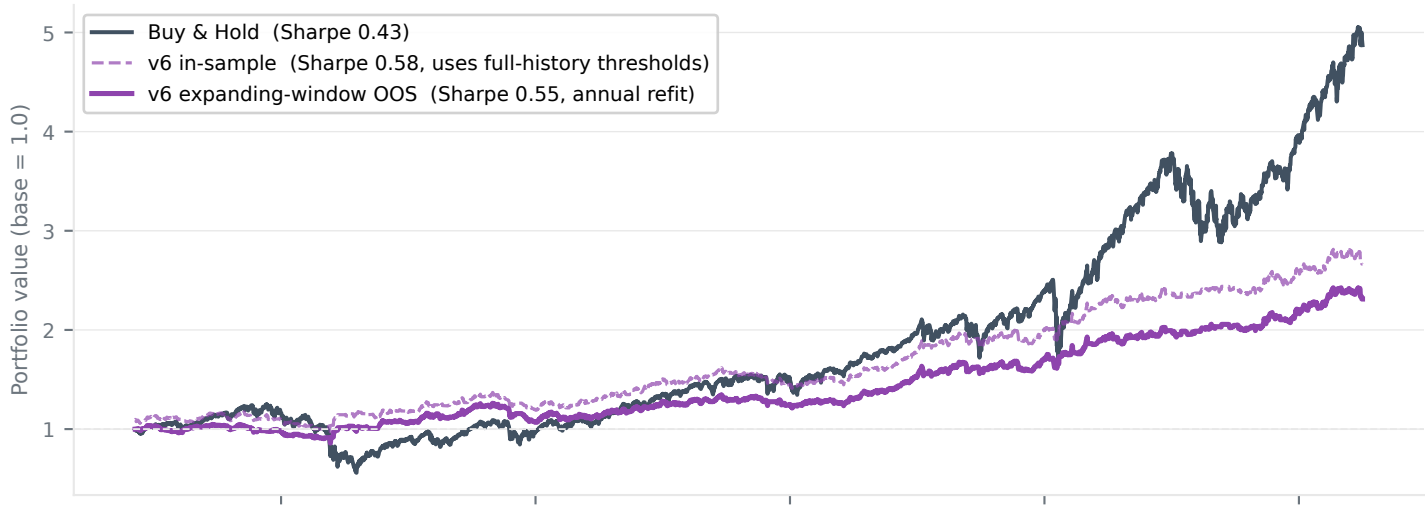
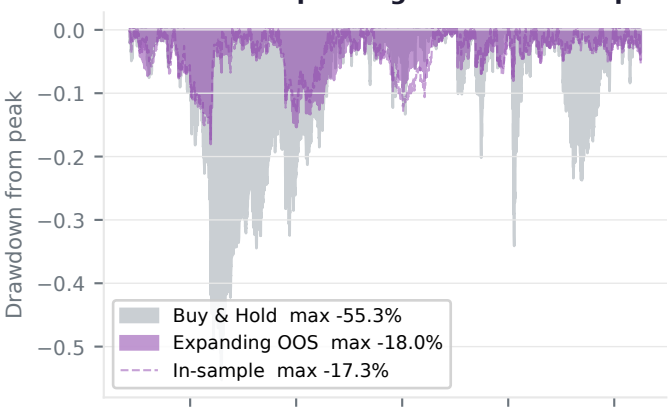
Expanding-Window OOS vs In-Sample (v6 multi-scale + vol ratio, 0 bps)

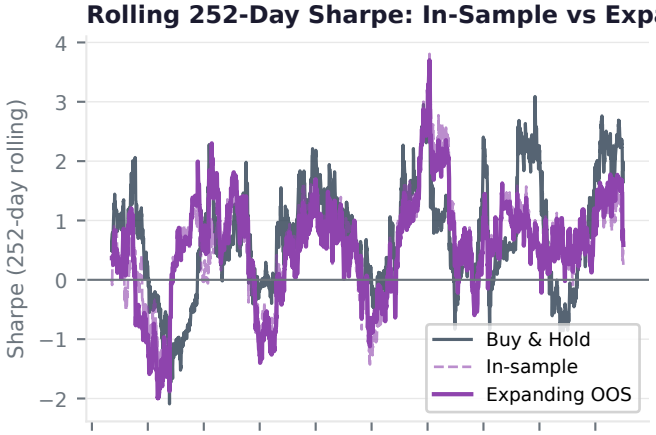


Expanding: geometric thresholds and Markov EM refitted annually on all prior data. Test period begins after 2-year minimum training window (~2002). Optimism bias = in-sample Sharpe (0.58) - expanding Sharpe (0.55) = +0.03.

Drawdown: Expanding OOS vs In-Sample



Rolling 252-Day Sharpe: In-Sample vs Expanding OOS



In-sample uses full-history thresholds throughout. Expanding OOS refits annually. Convergence of the two lines indicates low optimism bias in a given period.

Version History - Key Changes and Impact

Version	Key Change	Data	CAGR	Sharpe	T-stat sig.
v1.0	Geometric + Markov k=3 ensemble	SPY 2022-2025	--	--	--
v2.0	Walk-forward leakage fix; 2000-2025 data	SPY 2000-2025	--	--	--
v3.0	BIC extended; VIX pipeline	SPY 2000-2025	+1.8%	0.29	p=0.19
v4.0	Half-position mixed; persistence filter	SPY 2000-2025	+5.7%	0.68	p=0.002
v5.0	Walk-forward OOS curve; multi-asset validation	SPY+4 assets	+5.7%	0.68	p=0.002
v6.0	Vol ratio; multi-scale geo; expanding-window BT	SPY 2000-2025	+4.8%	0.58	p=0.008